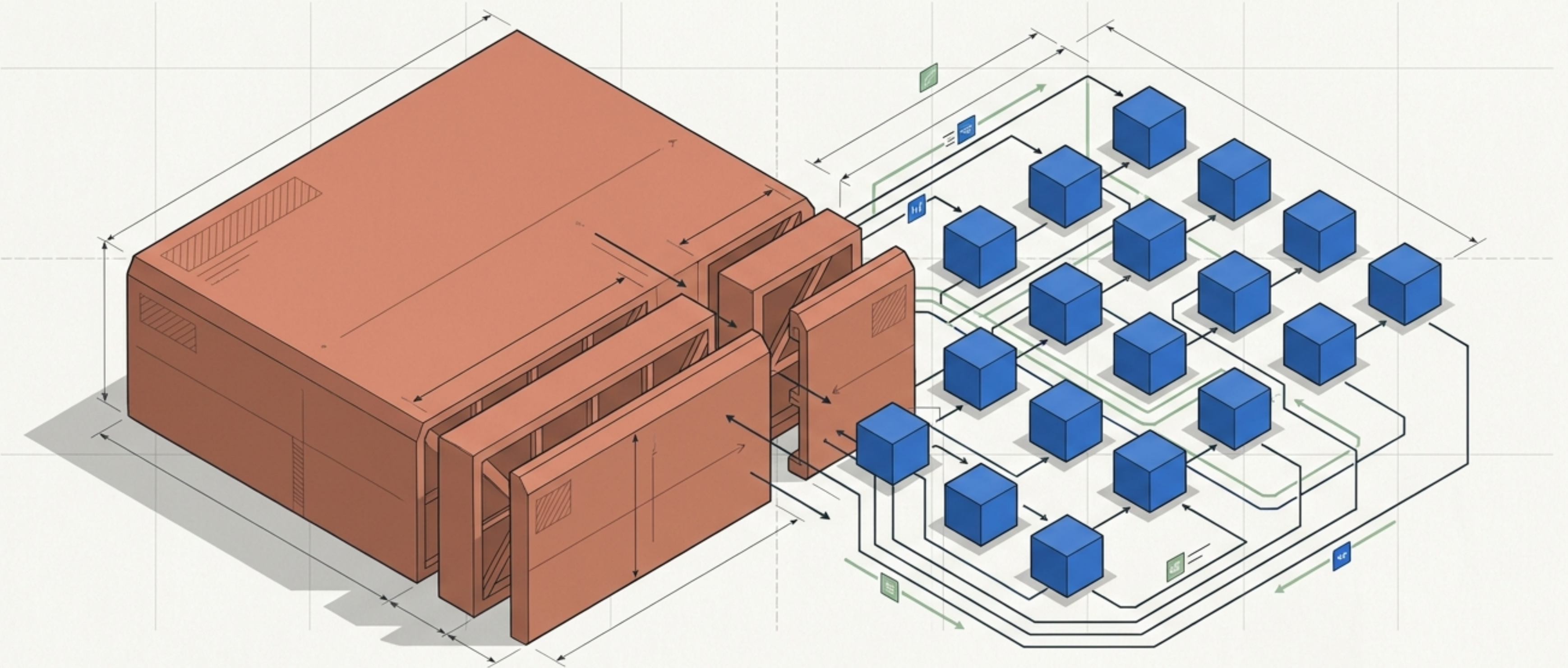


FROM MONOLITH TO MICROSERVICES

A Practical Guide to Aligning Strategy, Scaling Tech, and Accelerating Cloud Adoption via TOGAF Enterprise Architecture.
Based on the insights of Venkatesh R., Enterprise Architect.



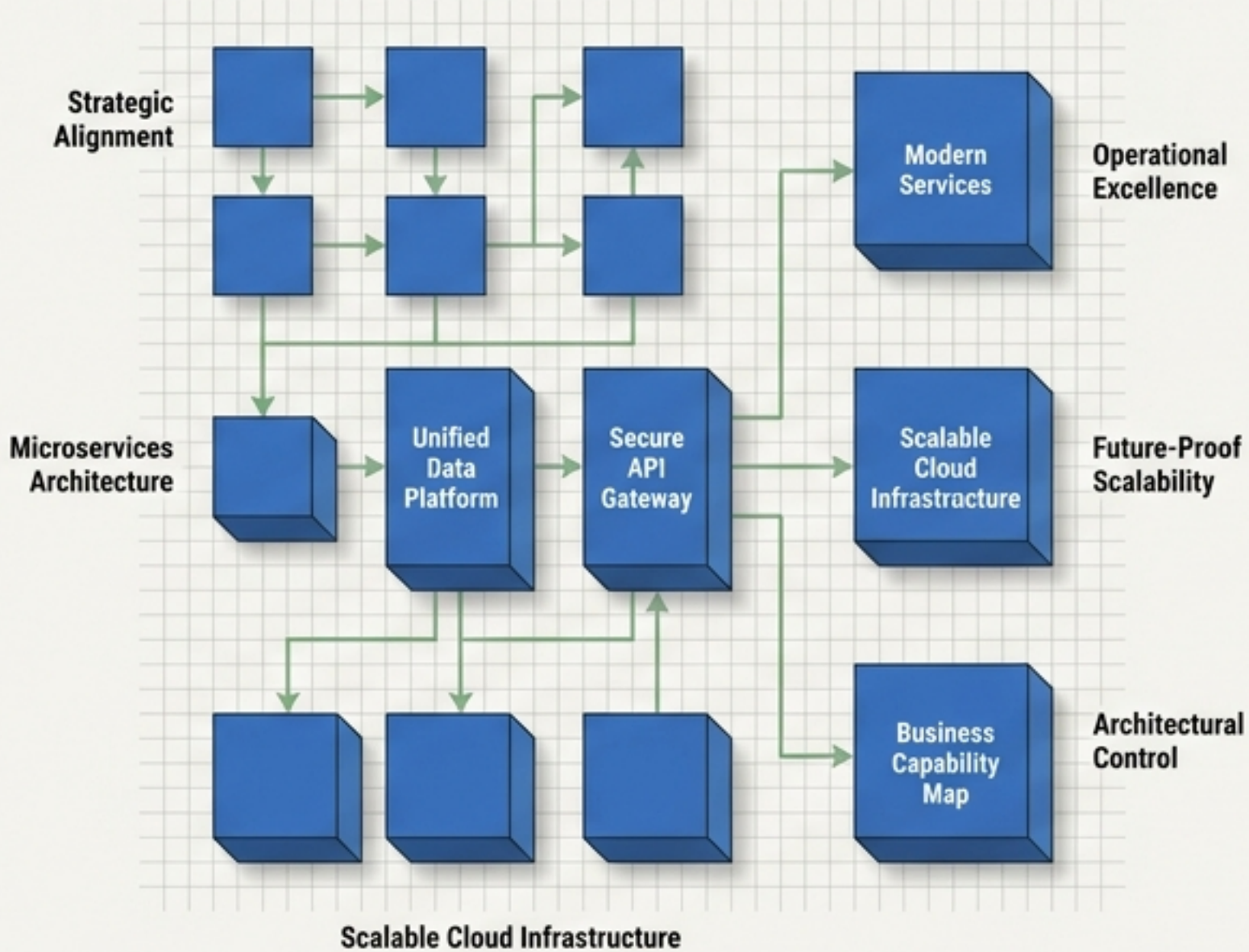
CLOUD MIGRATION IS A BUSINESS TRANSFORMATION, NOT JUST A TECH UPGRADE

Without a clear architectural vision, migrations collapse into misaligned priorities, security gaps, and integration chaos. Enterprise Architecture provides the structural compass.

AD-HOC MIGRATION: INTEGRATION CHAOS & TECH DEBT



TOGAF EA: STRATEGIC ALIGNMENT & SCALABILITY



EVERY ARCHITECTURAL ARTIFACT FITS INTO FOUR INTERCONNECTED DOMAINS

The TOGAF framework structures enterprise architecture into four foundational layers. Strategy dictates infrastructure.

Business Architecture

Strategy, organizational structure, core processes, business capabilities.

Data Architecture

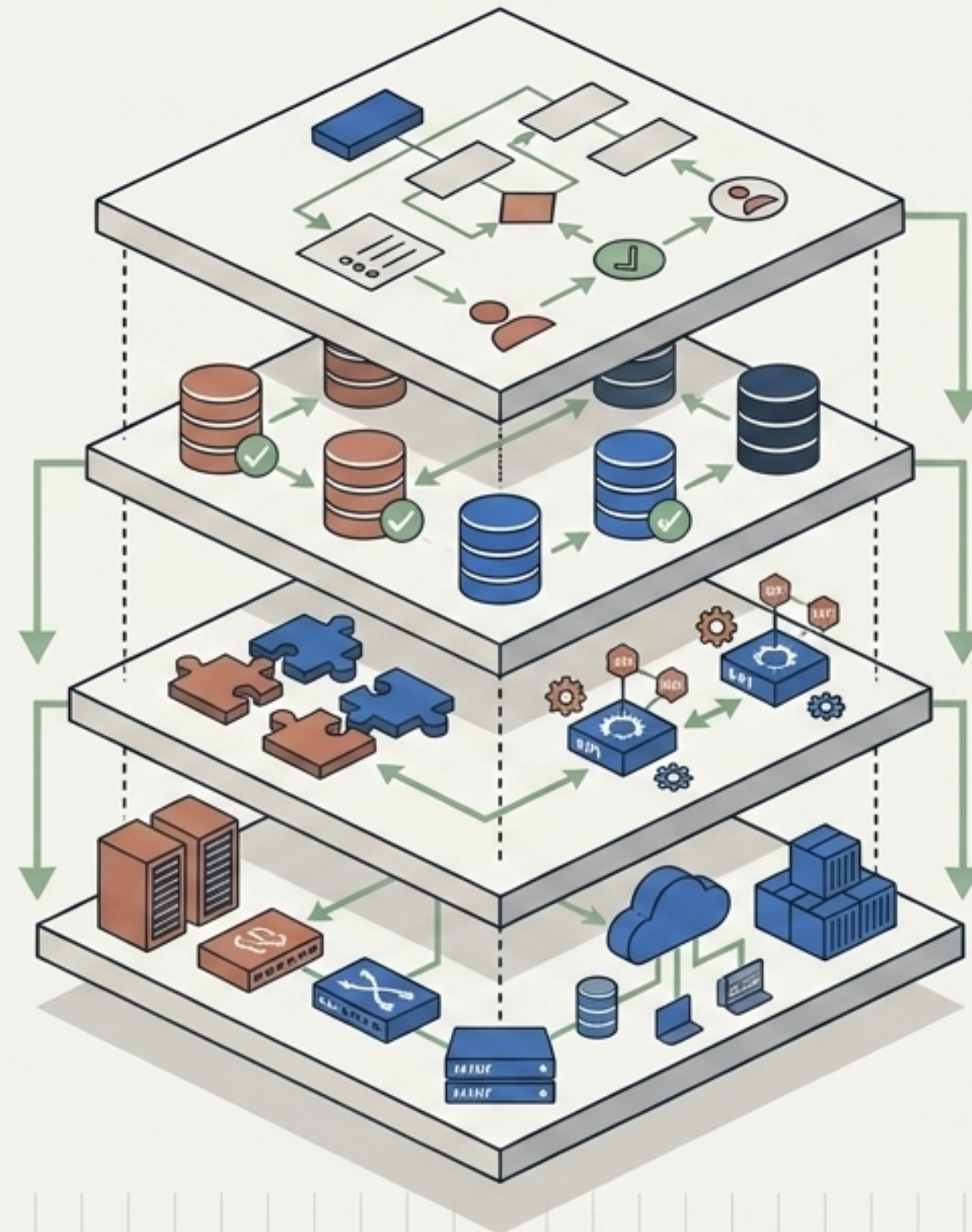
Data entities, data flows, master data management, data governance.

Application Architecture

Software systems, distributed services, APIs, system interactions.

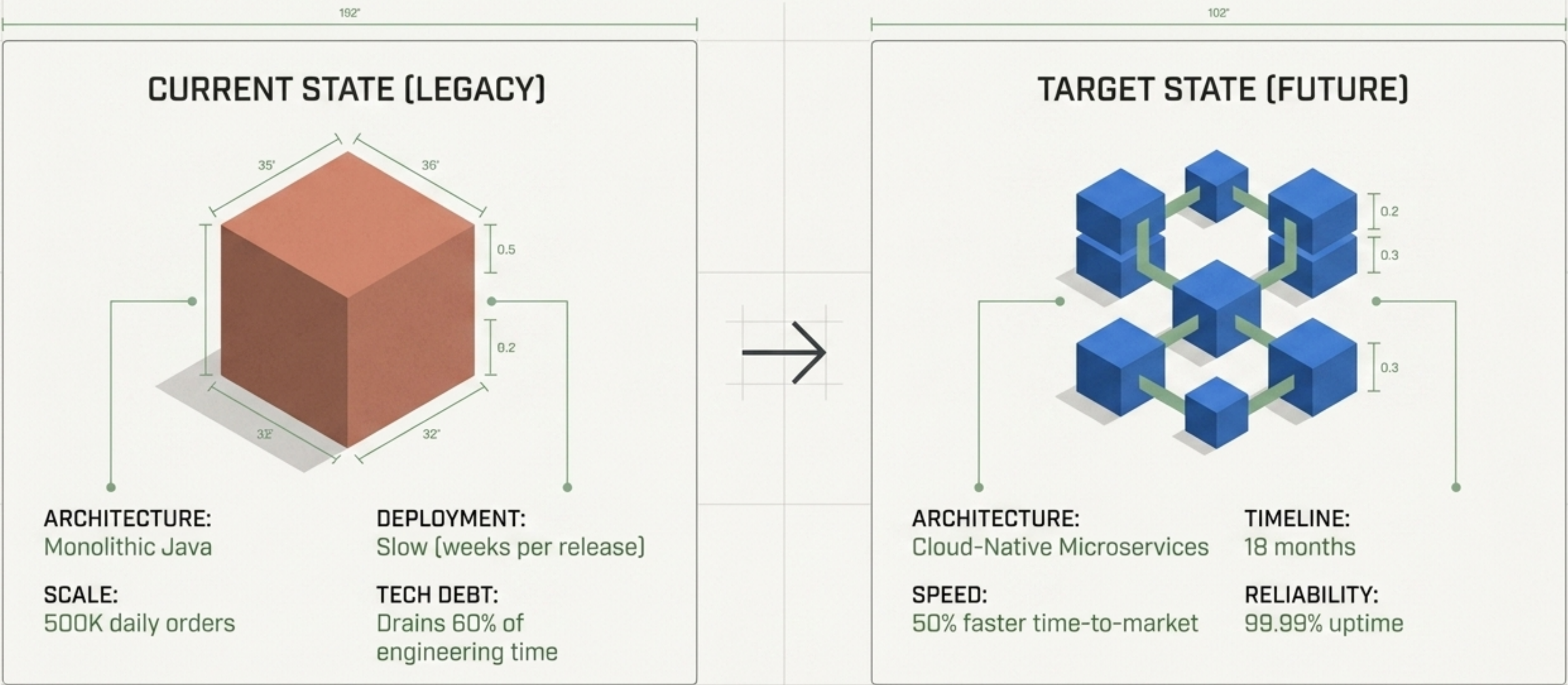
Technology Architecture

Hardware, networks, middleware, containers, infrastructure-as-code.



THE PROTAGONIST: SHOPFAST'S PLATFORM MODERNIZATION

ShopFast is a mid-sized e-commerce platform hitting the operational limits of its legacy architecture.

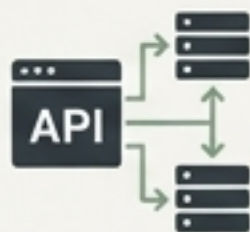


PHASE 0 & A: SETTING THE FOUNDATION AND THE NORTH STAR

PHASE 0: PRELIMINARY PRINCIPLES



Cloud-Native by Default:
Workloads run on Kubernetes.



API-First Architecture:
Versioned APIs built before UIs.



Event-Driven: Asynchronous communication via Kafka.



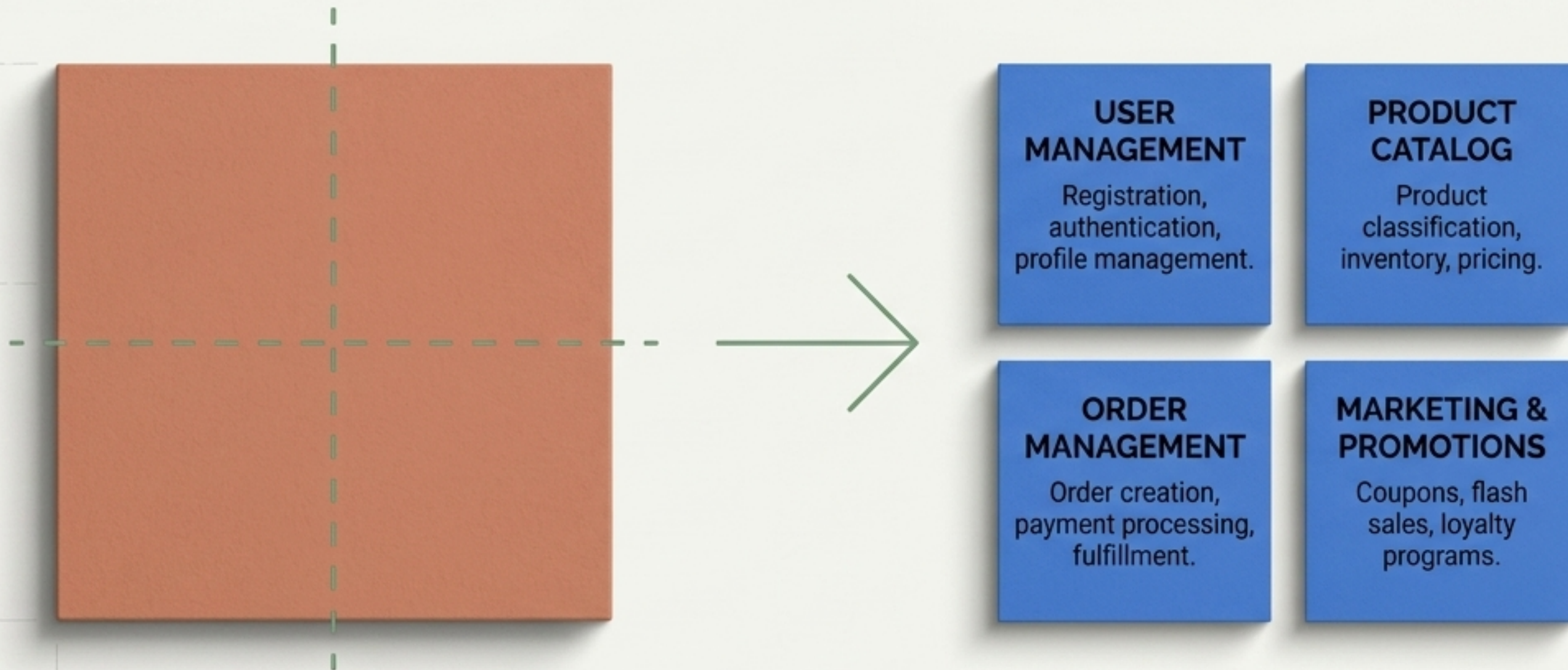
Observability Required: Structured logs, metrics, and traces.

PHASE A: ARCHITECTURE VISION

By **Q4 2026**, ShopFast will operate on a cloud-native microservices platform running on Kubernetes, with APIs enabling **10K transactions/second**, **99.99% uptime**, and the ability to deploy new features within **2 days**.

PHASE B: DECOMPOSING THE MONOLITH VIA BUSINESS CAPABILITIES

Business decomposition dictates microservice boundaries before a single line of code is written, ensuring strict adherence to the Single Responsibility Principle.



GAP ANALYSIS: Comparing the legacy monolith capabilities against this target state reveals the precise operational gaps required to modernize.

PHASE C: THE DATA DISTRIBUTION MATRIX

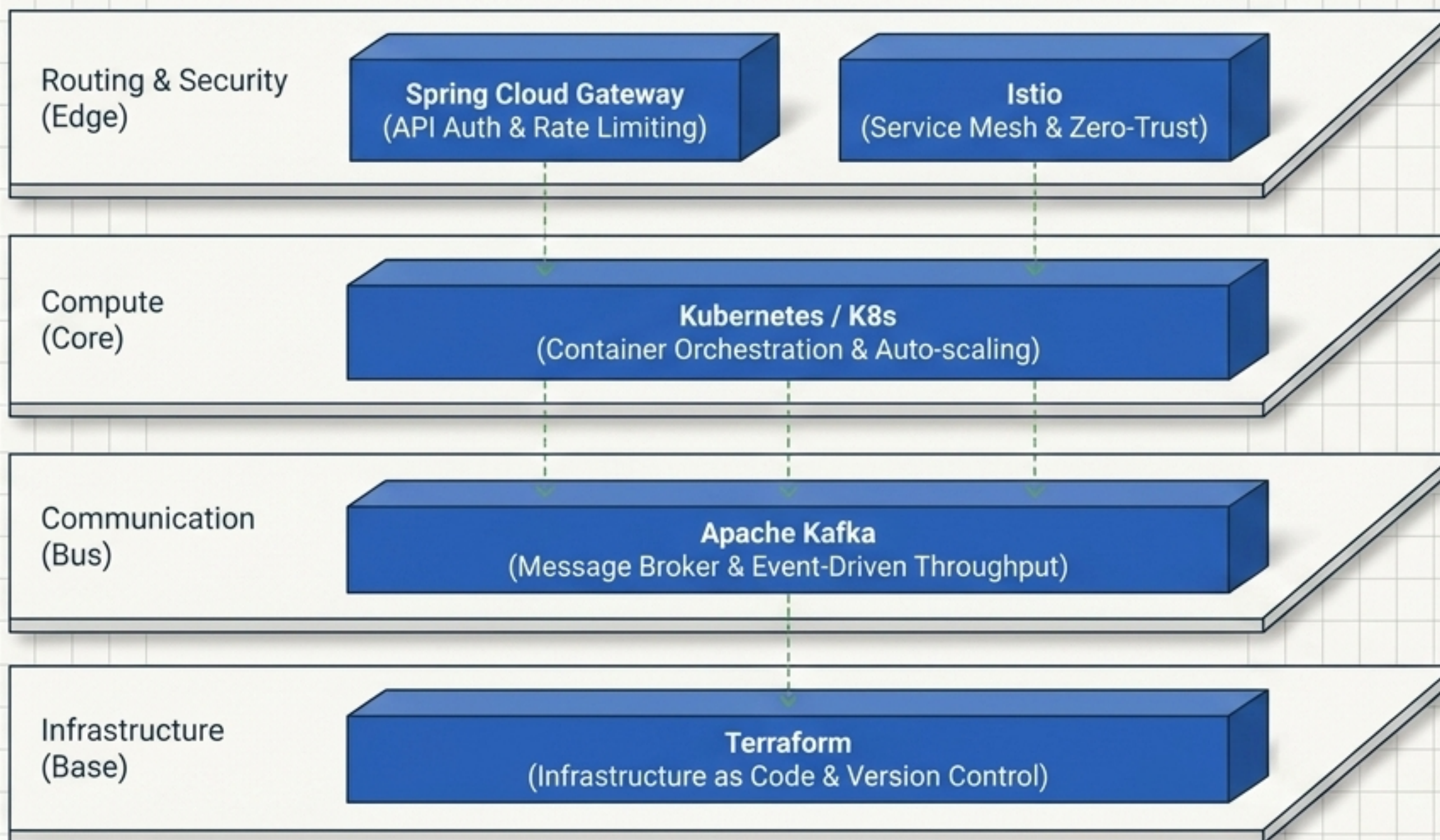
Defining the Information Systems Architecture. This critical artifact prevents tight coupling through shared databases by explicitly mapping Data Entities against Application Services.

	User Profile	Inventory	Order Status	Payment Ledger
User Service (Java)	[C]	[R]		
Product Service (Java)		[C]		
Order Service (Java)	[R]	[R]	[C]	[R]
Payment Service (Java)			[R]	[C]

[C] = Create/Own Domain Data | [R] = Read/Subscribe via **Event Sourcing** & Saga Patterns

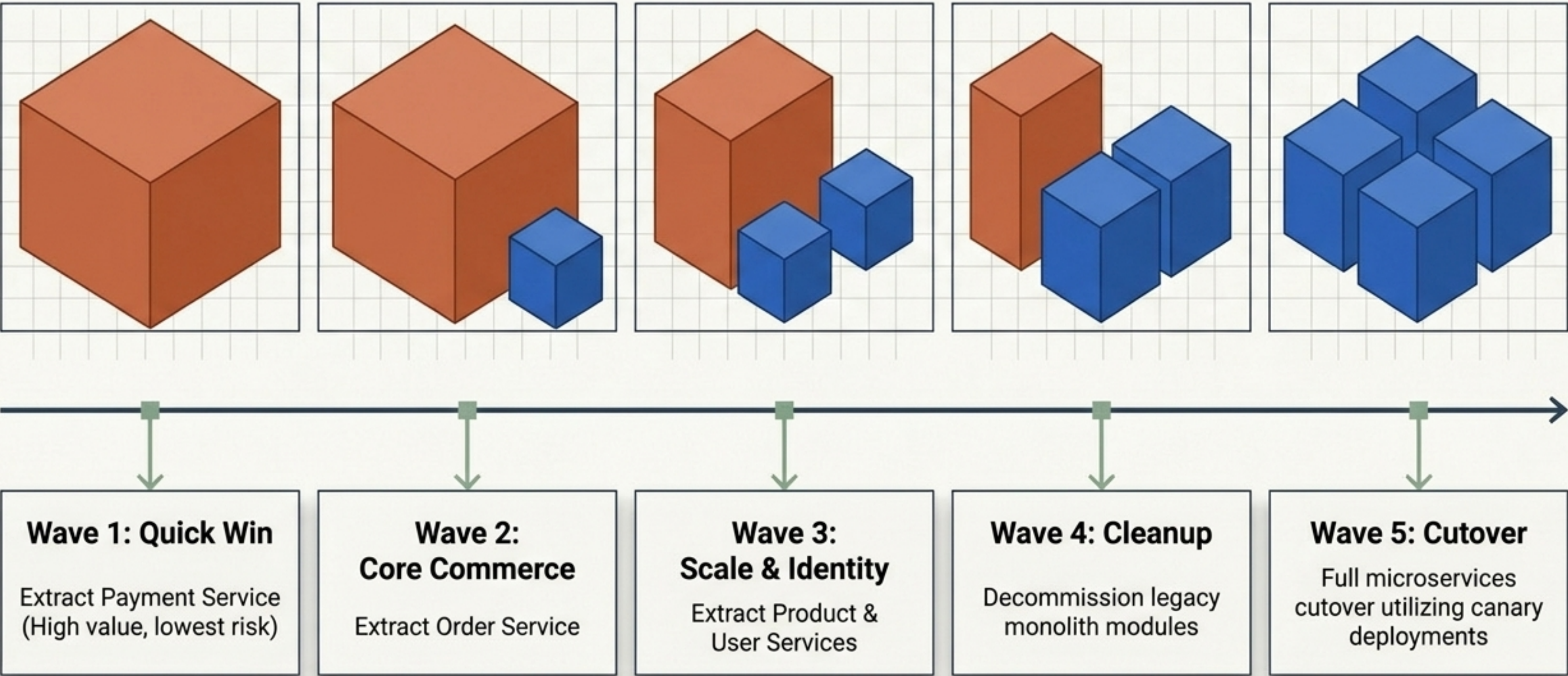
PHASE D: ARCHITECTING THE CLOUD-NATIVE FOUNDATION

Informed by the TOGAF Technical Reference Model, technical selections enforce Zero-Trust Security by Design across all layers.



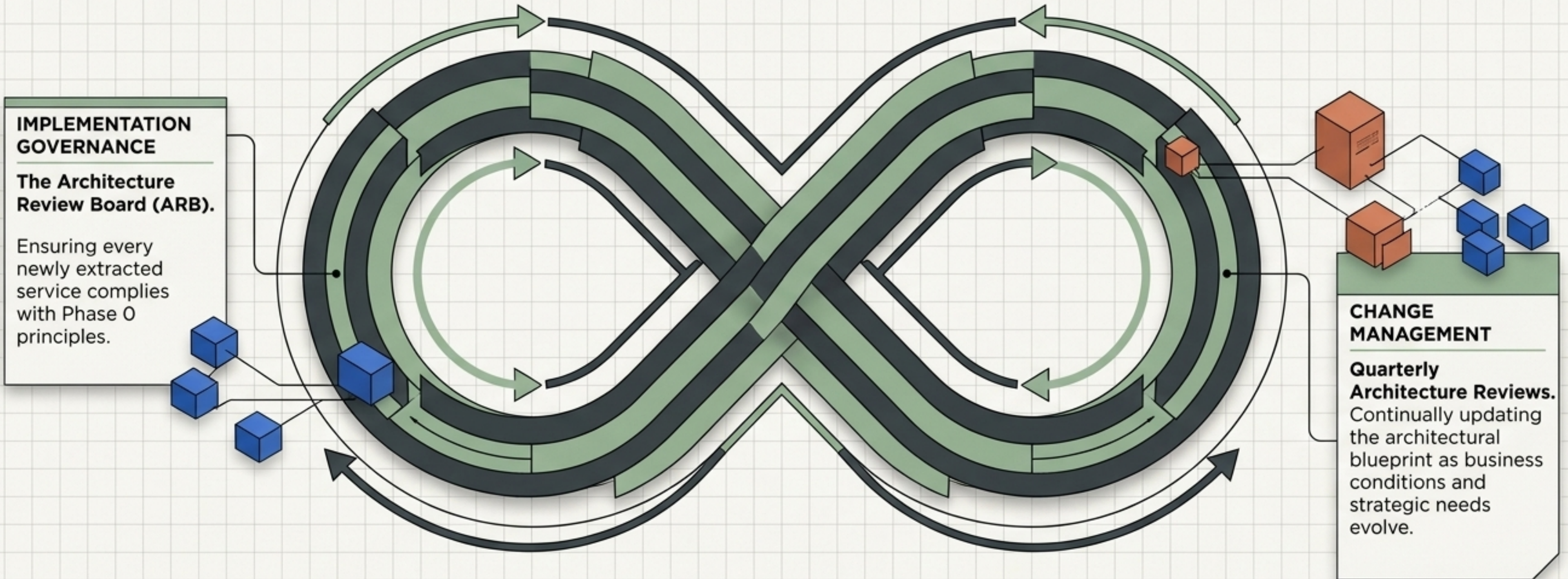
PHASE E & F: THE 5-WAVE MIGRATION ROADMAP

Grouping architectural gaps into defined work packages spanning 6-month value delivery milestones.



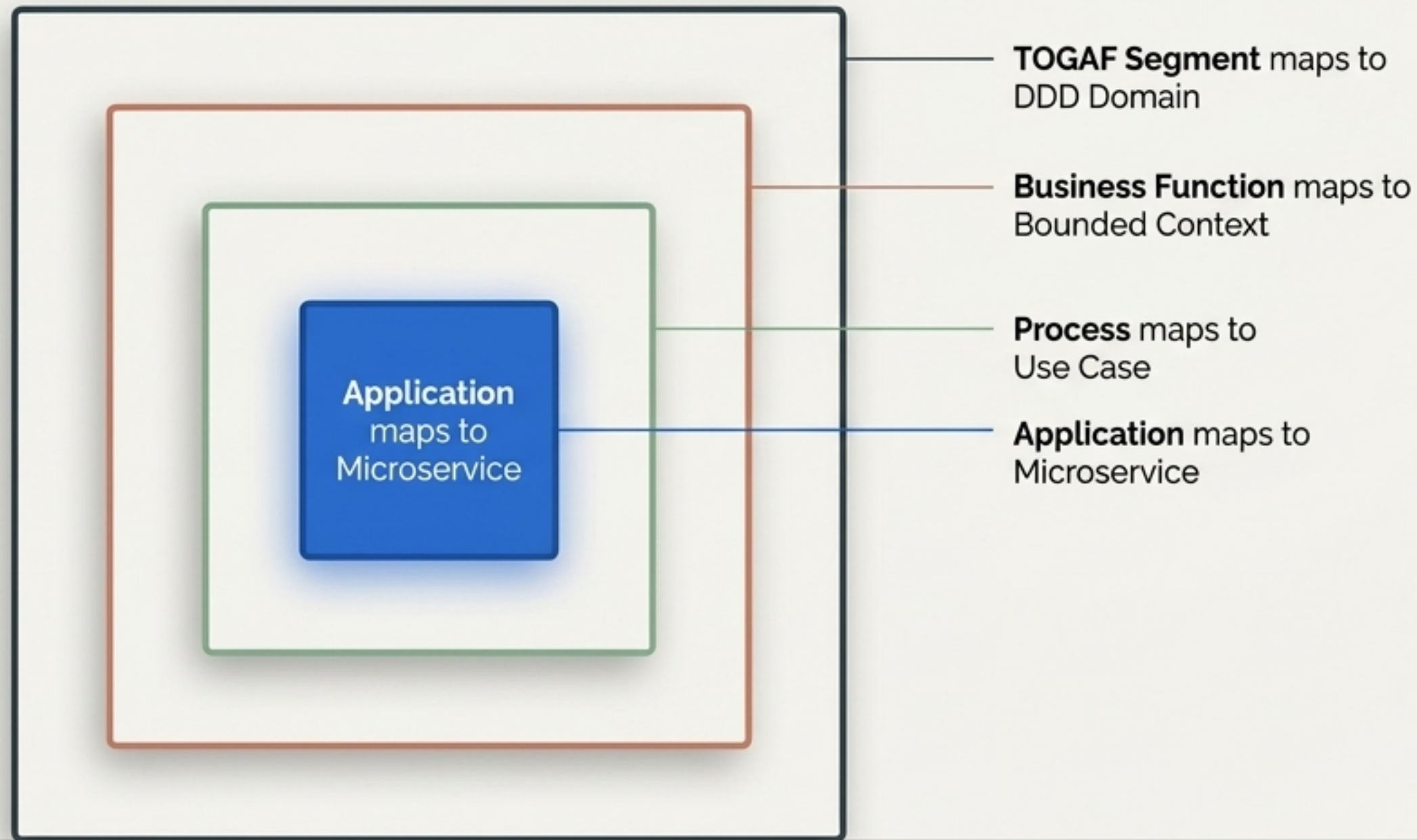
PHASE G & H: CONTINUOUS GOVERNANCE AND EVOLUTION

Architecture is never done. Governance is not about blocking development; it is about maintaining agility through consistent standards and principles.



ARCHITECTURE ACCELERATORS: BRIDGING TOGAF AND DOMAIN-DRIVEN DESIGN (DDD)

The TOGAF metamodel maps perfectly onto modern Domain-Driven Design principles, enabling seamless translation between Enterprise Architects and Software Engineers.



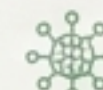
ADDITIONAL REFERENCE MODELS:



TOGAF TRM: Standard taxonomy for tech selection.



Cloud Ecosystem Model: Reusable Architecture Building Blocks (ABBs).



EIRA: European Interoperability Reference Architecture.

GLOBAL INDUSTRY VALIDATION: MEASURABLE TRANSFORMATION OUTCOMES

RETAIL (DFI RETAIL GROUP)



USE CASE: Omnichannel Cloud Migration.

OUTCOME:

Architecture docs completed 65% faster; implementation 4 months ahead of schedule. Decomposed to Spring Boot & GKE.

BANKING (RABOBANK / WESTPAC)



USE CASE: Core Banking & DevOps.

OUTCOME:

Embedded continuous EA governance, created distinct API layers, and achieved rapid legacy decommissioning via regulatory gates.

HEALTHCARE (JORDAN / BOTSWANA)



USE CASE: National eHealth & Virtual Clinics.

OUTCOME:

Navigated interoperability complexity to drastically improve patient handling, waiting times, and resource utilization.

MANUFACTURING (GLOBAL SMEs)



USE CASE: ERP & Industrial Digital Platforms.

OUTCOME:

Applied TOGAF 10 + SCOR-DS logic to achieve remote operational management and improved Overall Equipment Effectiveness (OEE).

THE ARCHITECT'S PLAYBOOK: YOUR 5-STEP ACTION PLAN

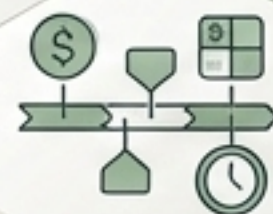
GOVERN, DON'T DICTATE.

Enforce via Compliance Checklists (e.g., OpenAPI specs) and quarterly check-ins.



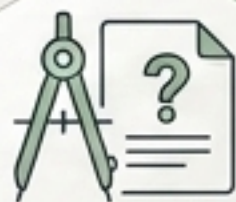
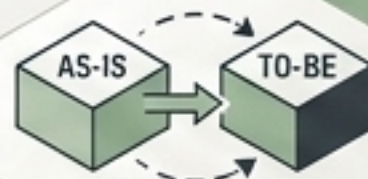
BUILD ROADMAP.

Group gaps into Work Packages. Prioritize based on Business Value vs. Effort.



MAP AS-IS VS. TO-BE.

Draw existing flowcharts against the ideal future state to explicitly list capability gaps.



START WITH WHY.

Draft a 1-page Vision Statement tying architecture directly to business metrics.

TAILOR THE FRAMEWORK.

Don't copy blindly. For a 6-month project, use only core phases and keep iterations short.



CORE ARCHITECTURAL PRINCIPLES

A TOOLKIT, NOT A RULEBOOK

Abbreviate and adapt TOGAF to your exact organizational context.



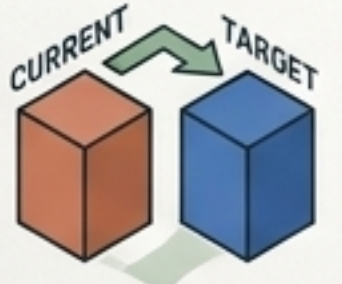
BUSINESS FIRST, TECH SECOND

Technology serves business capabilities. Never the other way around.



THE POWER OF GAP ANALYSIS

Comparing the Current state to the Target state is the most valuable step in revealing missing capabilities.



DOCUMENT THE WHY

Rigorously capture Architecture Decision Records (ADRs) to track the reasoning behind major choices.



COMPOSE IF ARCHITECTURE

Clear predefined architectural principles speed up—rather than slow down—engineering decision-making.

GOVERNANCE ENABLES AGILITY

Clear, predefined architectural principles speed up—rather than slow down—engineering decision-making.



WHERE WILL YOUR ARCHITECTURE TAKE YOU?

Master the framework, refine your strategy, and architect the future of your enterprise.

FURTHER READING

- The Open Group Case Studies (opengroup.org/case-studies)
- E-Commerce Architecture Blueprint (55 real diagrams in Sparx EA)
- Visual Paradigm TOGAF Guide (AI-enhanced ADM walkthroughs)
- TOGAF Standard, 10th Edition

CONTINUE THE CONVERSATION

Author: Venkatesh R. | Enterprise Architect

Specialization: Cloud-Native Transformations & Architecture Governance

Connect: Medium @tatankavenkat_19803

What integration challenges is your enterprise facing? Drop a comment on the article.